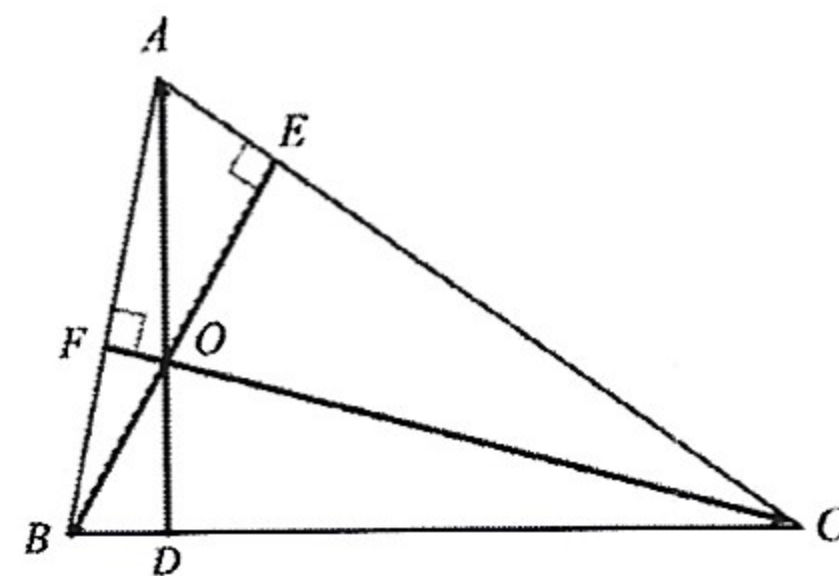


Vector WS 1

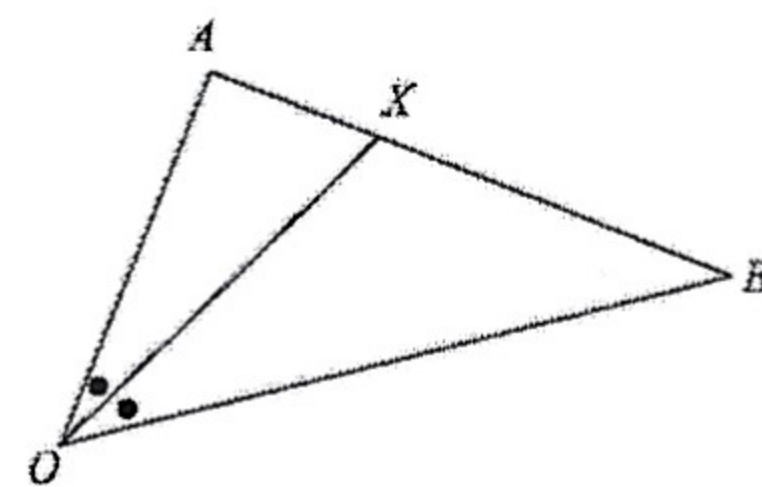
- $\triangle OAB$ is isosceles with $\vec{OA} = (3\hat{i} + 2\hat{j} + \sqrt{3}\hat{k})$, $\vec{OB} = \alpha\hat{i}$ ($\alpha > 0$) and $|\vec{OA}| = |\vec{OB}|$.
 - Find the value of α .
 - Find $\vec{OC}$, where C is the midpoint of the line segment AB .
 - Show that $\vec{OC}$ is perpendicular to $\vec{AB}$.

- Given vector $\hat{u} = 3\hat{i} + m\hat{j} + \hat{k}$ where m is a real number. Find the value(s) of m :
 - If the length of vector $\hat{u}$ is 10.
 - If the vector $\hat{u}$ makes an angle of $\cos^{-1}(\frac{1}{3})$ with y axis.

- ABC is an acute angled triangle. The altitudes BE and CF intersect at O . The line AO produced meets BC at D . Relative to O the position vectors of A, B and C are $\hat{a}, \hat{b}$ and $\hat{c}$ respectively.
 - Show that $\hat{b} \cdot (\hat{c} - \hat{a}) = 0$ and $\hat{c} \cdot (\hat{b} - \hat{a}) = 0$.
 - Hence show that $AD \perp BC$.
 - What geometrical property of the triangle has been proved?

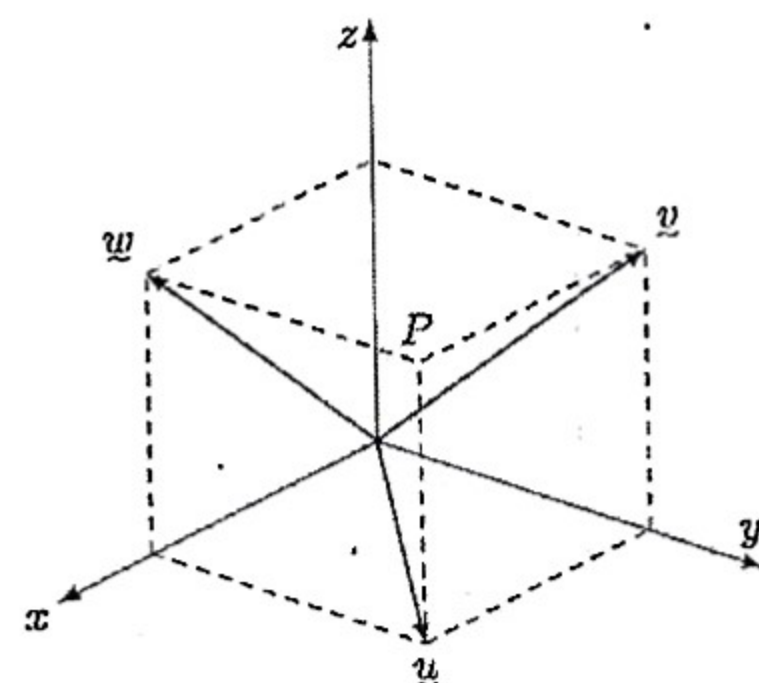


- Consider two points A and B whose position vectors are $\hat{a}$ and $\hat{b}$ respectively. The angle bisector of $\angle AOB$ meets AB at X .
 - Explain why $\vec{OX} = \lambda(\hat{a} + \hat{b})$ where $\hat{a}$ and $\hat{b}$ are the corresponding unit vectors for $\hat{a}$ and $\hat{b}$.
 - Let $\vec{AX} = \mu\vec{AB}$, where μ is a scalar and $0 < \mu < 1$. Use vector methods to show that OX divides AB in the ratio of the lengths of the adjacent sides i.e. $AX:XB = OA:OB$.



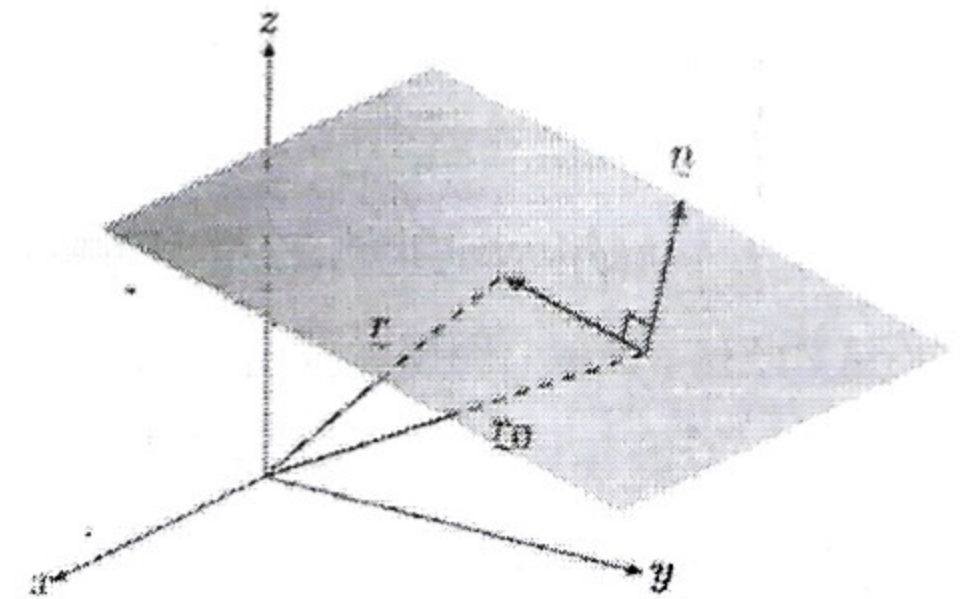
- If $P = i + j + k$ and $R = 9i + 3j + 8k$, find the point Q on $\vec{PR}$ such that $PQ:QR = 2:3$.
- If $\vec{u}$ and $\vec{v}$ are vectors defined by $\vec{u} = \hat{i} + \hat{j} + \sqrt{2}\hat{k}$ and $\vec{v} = \hat{i} - \hat{j} + \sqrt{2}\hat{k}$, find:
 - A unit vector parallel to $\vec{u}$
 - The angle between $\vec{u}$ and $\vec{v}$
 - The vector projection of $\vec{v}$ in the direction of $\vec{u}$
- For the curves whose vector equations are given, find the parametric equation and the Cartesian equation.
 - $\vec{r} = t\hat{i} + 4at^2\hat{j}, t \geq 0$
 - $\vec{r} = (2 - \sin \theta)\hat{i} + (1 + \cos \theta)\hat{j}, 0 \leq \theta \leq 2\pi$
- Graph the curve represented by each vector function.
 - $\begin{pmatrix} 1 \\ t \\ 1 \end{pmatrix}$ for $0 \leq t \leq 10$
 - $\begin{pmatrix} 1 \\ t \\ t \end{pmatrix}$ for $0 \leq t \leq 5$
 - $\begin{pmatrix} t \\ t^2 \\ 1 \end{pmatrix}$ for $0 \leq t \leq 3$

- The diagram below shows vectors $\vec{u}, \vec{v}$ and $\vec{w}$ in a rectangular prism produced by the origin and $P(a, b, c)$. Show that if the angles between $\vec{u}, \vec{v}$ and $\vec{w}$ are all equal to $\frac{\pi}{3}$, then the rectangular prism is a cube.



10. A plane in 3D space can be defined by a point on the plane and a vector perpendicular to the plane called the normal vector. This works similarly to how a line in 2D space can be defined by a point on the line and a gradient.

The diagram above shows two vectors $\vec{r}$ and $\vec{r}_0$ representing the position vectors of points $P(x, y, z)$ and $P_0(x_0, y_0, z_0)$ on a plane. Let $\vec{n} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ be the vector perpendicular to the plane.



- Explain why $(\vec{r} - \vec{r}_0) \cdot \vec{n} = 0$.
 - Hence, show that the equation of the plane is $ax + by + cz = ax_0 + by_0 + cz_0$
 - Find the equation of the plane that passes through $(2, -3, 1)$ and is perpendicular to the vector $-3\vec{i} + 2\vec{j} - \vec{k}$.
 - What is the shortest possible distance between any point on the plane $ax + by + cz + d = 0$ and the origin?
11. Two particles travel according to the trajectories defined by

$$\vec{r}_1(t) = (2t - 1)\vec{i} + (t^2)\vec{j} + (t + 1)\vec{k}$$

$$\vec{r}_2(t) = (t^2)\vec{i} + (3t - 2)\vec{j} + (3 - t)\vec{k}$$

Determine if the particles will ever collide.

12. Consider the two spheres

$$S_1: (x - 1)^2 + (y + 1)^2 + (z - 2)^2 = 16$$

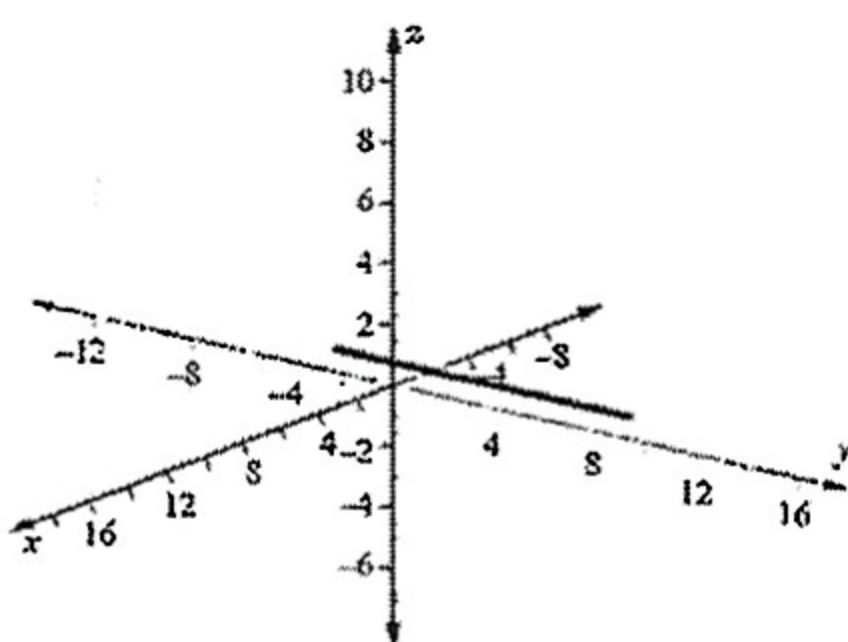
$$S_2: (x + 1)^2 + (y - 3)^2 + (z + 2)^2 = 4$$

- Show that S_1 and S_2 are tangential to each other
- Find the coordinates of the common point
- Find another possible radius for S_2 such that the spheres will be tangential
- Find the coordinates of the common point when this occurs.

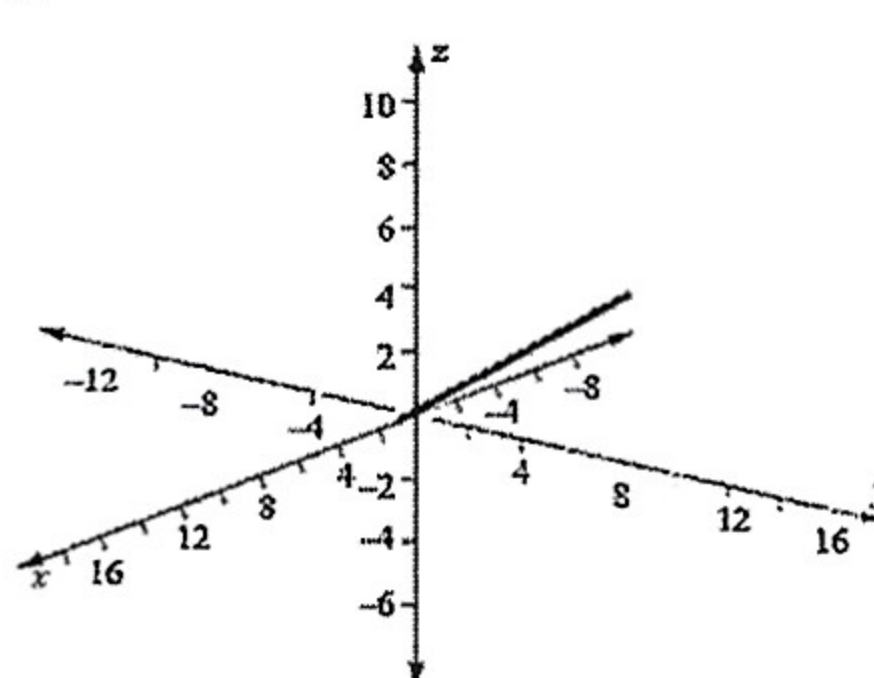
Answer

- a. 4 b. $\vec{OC} = \frac{1}{2}(7\vec{i} + 2\vec{j} + \sqrt{3}\vec{k})$
- a. $\pm 3\sqrt{10}$; b. $\pm \frac{\sqrt{5}}{2}$
- $(\frac{21}{5}, \frac{9}{5}, \frac{19}{5})$
- a. $\frac{1}{2}(\vec{i} + \vec{j} + \sqrt{2}\vec{k})$; b. 60° ; c. $\frac{1}{2}(\vec{i} + \vec{j} + \sqrt{2}\vec{k})$
- a. Parametric: $x = t, y = 4at^2, t \geq 0$, Cartesian: $y = 4ax^2, x \geq 0$; b. Parametric: $x = 2 - \sin \theta, y = 1 + \cos \theta, 0 \leq \theta \leq 2\pi$, Cartesian: $(x - 2)^2 + (y - 1)^2 = 1$
- 8.

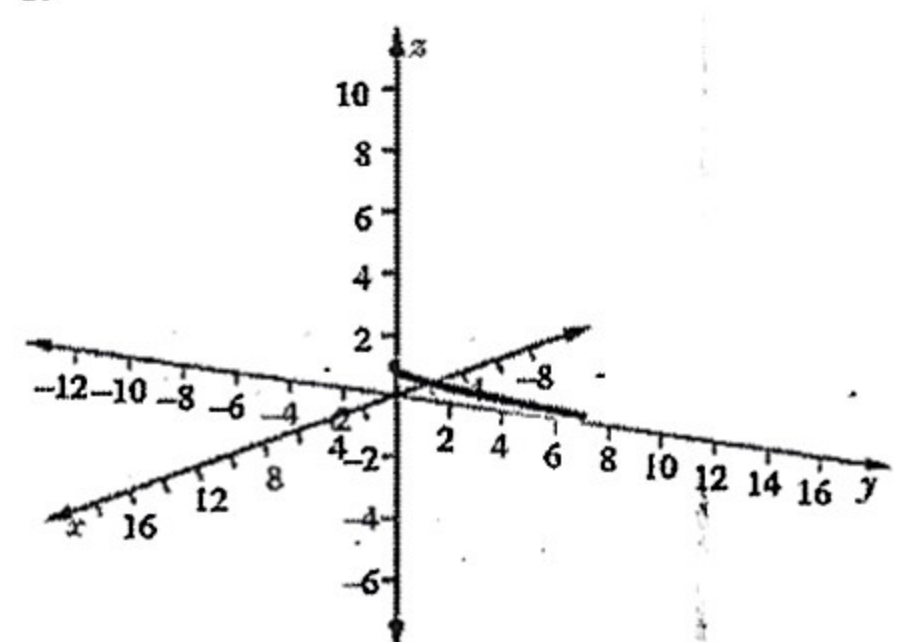
a.



b.



c.



- a. $(\vec{r} - \vec{r}_0)$ lies on the plane and $\vec{n}$ is perpendicular to the plane, which means that it is perpendicular to any vector on the plane. So their dot product is zero; b. Expand $(\vec{r} - \vec{r}_0) \cdot \vec{n} = 0$; c. $3x - 2y + z = 13$; d. $\sqrt{a^2 + b^2 + c^2}$
- The particles will collide when $t = 1$. This is found by equating any component, solving for t , and then testing the other components
- b. $(-\frac{1}{3}, \frac{5}{3}, -\frac{2}{3})$; c. 10; d. $(\frac{7}{3}, -\frac{11}{3}, \frac{14}{3})$